Example: The water jugs problem:

Given: 2 jugs:

Problem: fill the 4 l jug with 2 l of water.

Representation:

- a state:
  - [content of large jug, content of small jug]
- initial state:
  - [0, 0]
- goal state:
  - [2, x]

Rules for the jugs example

- <u>Fill large:</u>

  [x, y] and x < 4 → [4, y]

- <u>Fill small:</u>

  [x, y] and y < 3 → [x, 3]

- <u>Empty large:</u>

  [x, y] and x > 0 → [0, y]

- <u>Remove some from large:</u>

  [x, y] and x > d > 0 → [x - d, y]

- <u>Empty (remove some from) small.</u>